Quant Week 4
2/20/25

Agenda

1. Icebreaker Question
2. Brainteasers
3. Greek Options Values
4. Project

Ice Breaker

Favorite flavor (of anything)

Brain Teaser - Bankers

2 bankers arrive at the station at some random time between 5 and 6am. (uniform distribution)

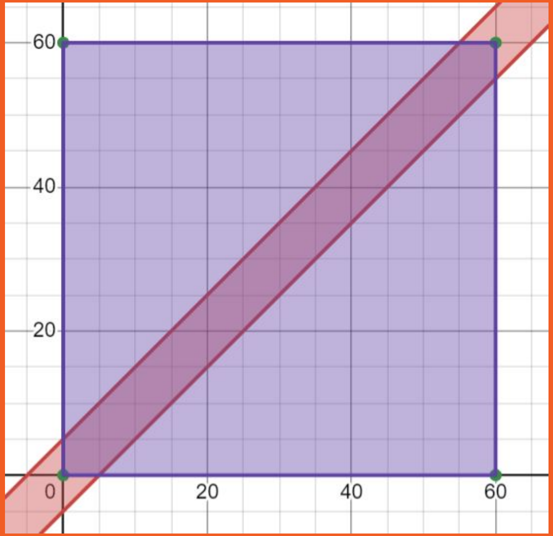
They stay exactly 5 minutes then leave.

On a given day, what is the probability they will run into each other?

Brain Teaser - Bankers - Solution

Let’s say these people are person X and person Y. Their arrival times are each between 0 and 60.
They will meet when $|X - Y| \leq 5$

Find the area of the red strip
(60*60 - 2*(½*55*55)) / 3600
= 23/144



Technical formulaic way of thinking about this problem: where f=1/3600

Hint: If $f_{X,Y}$ is the multivariate pdf then you want to solve the following integral

$$\int_0^{60} \int_{\max\{0,x-5\}}^{\min\{60,x+5\}} f_{X,Y}(x,y)\,dy\,dx.$$

Option Greek Values

What are the Greeks?

The Greek values are metrics that are used to determine the performance of options in relation to their underlying asset.

These are the first partial derivatives of various factors influencing the asset value.

The primary greek values are:

Delta Δ, Theta Θ, Gamma Γ, Vega ν, and Rho ρ

Note: **Alpha** and **Beta** are not included here.

Alpha (not a Greek)

This measures how good/bad a stock has performed compared to a benchmark

Often measured in percentage

Example:

Stock Starting Price: $20
Stock Ending Price: $21
Dividend: $1
Alpha = (21 + 1 - 20) / 20 = 10%

From Investopedia:

Formula for Alpha:

$$\text{Alpha} = \frac{\text{End Price} + \text{DPS} - \text{Start Price}}{\text{Start Price}}$$

where:

$\text{DPS} = \text{Distribution per share}$

Beta β (not a Greek)

$$\text{Beta coefficient}(\beta) = \frac{\text{Covariance}(R_e, R_m)}{\text{Variance}(R_m)}$$

Even though Beta is a greek letter, it is not considered a greek.

It doesn't represent a partial derivative of the underlying asset like the other values do.

Beta represents the volatility of an asset compared to the rest of the market.

In terms of statistics, this is very similar to variance but is normalized to the market as a whole

A beta of greater than 1 means the asset is more volatile than the market
A beta of less than 1 means the asset is less volatile than the market

Look at NVDA

NasdaqGS - Nasdaq Real Time Price • USD
NVIDIA Corporation (NVDA)
Follow
Compare
Time to buy NVDA?
138.85 +3.56 +(2.63%)
At close: February 14 at 4:00:01 PM EST
138.46 -0.39 (-0.28%)
After hours: February 14 at 7:59:56 PM EST
1D 5D 1M 6M YTD 1Y 5Y All
Key Events
139.00
138.46
138.00
137.00
136.00
135.29
06:00
08:00
10:00
12:00
14:00
16:00
18:00
2/14 13:02

Delta Δ

Estimates the change in price of the derivative/option based on a $1 move in the price of the underlying asset

This value is typically not above 1.0

Example,

A stock price increases by $2.

A call option with a strike price of $80 that is currently trading at $20 has a delta of 0.8 What will the new price of this be?

$21.60

Option Chain - Strike Price

CALLS								PUTS						
IMP VOLATILITY	VOLUME	OPEN INT	THETA	DELTA	BID	ASK	STRIKE	BID	ASK	DELTA	THETA	OPEN INT	VOLUME	IMP VOLATILITY
59.64%	0	6	-0.02	0.97	31.20	31.40	70	0.13	0.14	-0.02	-0.01	226	33	54.12%
53.47%	0	6	-0.02	0.96	26.25	26.50	75	0.22	0.24	-0.03	-0.01	3,258	21	49.73%
48.66%	1	72	-0.03	0.93	21.50	21.70	80	0.41	0.43	-0.06	-0.02	6,773	27	45.97%
44.42%	6	315	-0.03	0.89	16.85	17.05	85	0.79	0.81	-0.11	-0.03	9,664	77	43.10%
42.21%	8	495	-0.05	0.81	12.60	12.80	90	1.49	1.52	-0.18	-0.04	2,929	693	40.86%
39.98%	276	5,557	-0.06	0.70	8.85	9.00	95	2.72	2.76	-0.30	-0.05	5,153	56	39.28%
38.48%	633	16,235	-0.06	0.56	5.85	5.90	100	4.60	4.70	-0.44	-0.06	15,850	532	38.15%
37.54%	781	7,833	-0.06	0.41	3.50	3.60	105	7.30	7.45	-0.59	-0.06	381	88	37.59%
37.00%	3,737	4,307	-0.05	0.28	2.02	2.05	110	10.80	10.90	-0.72	-0.05	283	1	37.05%
36.74%	72	4,713	-0.04	0.17	1.09	1.10	115	14.80	15.00	-0.82	-0.04	9	0	37.38%
37.15%	13	892	-0.03	0.10	0.56	0.59	120	19.30	19.50	-0.89	-0.03	16	0	38.22%
37.64%	7	291	-0.02	0.06	0.30	0.31	125	24.00	24.25	-0.93	-0.02	11	0	39.89%
38.90%	3	179	-0.01	0.04	0.16	0.18	130	28.90	29.10	-0.95	-0.02	2	0	41.50%

Theta Θ

Estimates the rate of change between price and time

Value will get larger as the option gets closer to date of expiration

In-the-money options will have a positive theta value, out of the money options will be negative

Example,

A theta value is -0.5

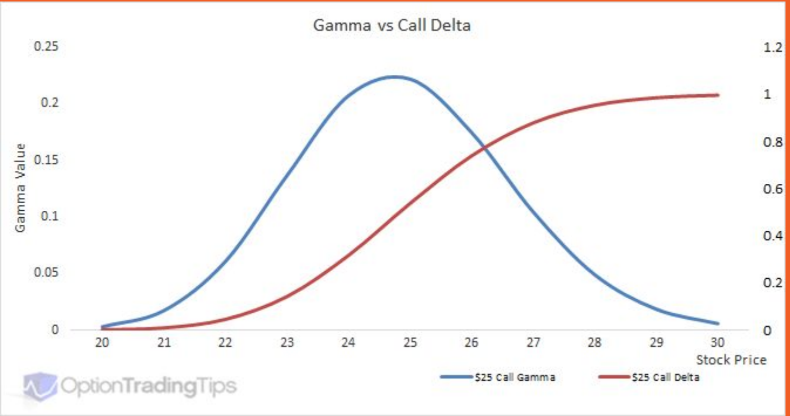
This means that for every day that passes, the price of the option will go down by $0.50

Gamma Γ

The derivative of Delta with respect to price

Basically how fast the Delta is changing at a price

Higher gamma values mean more risk. Given 2 stocks with equal delta values that each have price changes of $5, the one with a higher gamma will have a larger change in price



Vega v

Measures change in price based on change in implied volatility.

Forward-looking

Whereas delta measures in terms of actual price change, Vega measures in terms of theoretical price changes

Rho ρ

Measures Price change in terms of Interest Rates

Other Lesser Known Greeks

Lambda, Epsilon, Vomma, Vera, Zomma, and Ultima

- These are second and third derivatives of the other greeks we have talked about
- These trends can also be seen by those with a keen eye by looking at the main Greeks, but are explicitly calculated and named to be these variables
- Becoming increasingly more important in large algorithms because they are relatively easy to compute and can show hidden patterns

Project Discussion/Work Time